

Oil Production Prediction from Wellhead Data Using Machine Learning

Chinaecherem Anyanwu – CHE 696 Final Project

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1. Summary

This study develops a data-driven framework for predicting oil production from wellhead data using non-linear ensemble models. The approach follows a structured five-phase workflow, including feature engineering and nested cross-validation. This ensures reliable and unbiased model evaluation. Results show that CatBoost achieves the best performance ($R^2 \approx 0.903$), with XGBoost and Random Forest also demonstrating strong predictive capability. Simpler models, however, fail to capture the system's complexity. Exploratory diagnostics and feature analysis consistently identify key drivers such as water cut (bsw_pct) and choke size (choke_64), along with fthp_psig and water_bpd. These trends reflect the non-linear structure of the production system and are consistent with past research and field observations. Targeted feature selection improves generalization and outperforms PCA, KPCA, and raw datasets. Overall, the framework provides accurate and reliable production predictions with applications in oil production optimization and well performance analysis.

Key words: Oil production; Wellhead data; Feature selection; Ensembles learning

2. Introduction

Crude oil remains the foundation of the global energy and industrial economy [1], with demand exceeding 106.5 million barrels per day (BPD) in 2026 [3]. For major producing nations like Nigeria, which ranks among the top global producers, accurate production forecasting is the backbone of oilfield optimization. These predictions are vital for reducing operational uncertainty, which can cost offshore operations more than \$35 million annually due to unplanned downtime and suboptimal resource management [4]. Therefore, developing tools that can reliably transform raw wellhead measurements into actionable production insights is a critical engineering priority.

Since the mid-20th century, the petroleum industry has relied heavily on empirical formulas, most notably the popular Gilbert correlation (1954), to estimate production rates [1][2]. While the Gilbert equation offers simplicity and cost-efficiency, it was developed using a limited dataset from California and often fails to capture the non-linear patterns and noise inherent in modern multiphase flow across regions [1]. More recent efforts to refine these estimates in Nigerian fields have utilized General Reduced Gradient optimization within tools like Excel Solver [2]. However, these methods are prone to linearization errors, sensitive to initial guesses, and exhibit poor scalability when faced with the "big data" now available in digital oilfields. Machine learning (ML) addresses this by learning complex nonlinear relationships directly from data without relying on simplifying assumptions about multiphase flow [1][2].

The primary objective of this project is to bridge this gap by building a data-driven tool that accurately predicts oil production rates (BPD) from surface wellhead data while prioritizing ML model generalization. Unlike complex, time-consuming physical simulations, machine learning (ML) offers a scalable and fast alternative [2]. It leverages historical data to predict performance for new wells with no production history. This study specifically addresses whether ML can identify well features that are consistent with the process physics and overcome the limitations of conventional empirical methods to provide unbiased production estimates.

<i>Wellhead data</i>	<i>Name</i>	<i>Description</i>
<i>Features</i>	Choke size (choke_64)	Controls how much fluid leaves the well
	Flow tubing head pressure (fthp_psig)	Pressure that drives fluids to the surface
	Basic sediment & water (bsw_pct)	Water and sediments (oil impurities) mixed with oil
	Water (water_bpd)	Amount of water produced (with oil) per day
	Separator temperature (sep_tem_F)	Temperature affecting phase separation efficiency
<i>Target</i>	Crude oil production (Oil (BPD))	Daily oil output from the well

Table 1: Physical description of selected wellhead features and target, reflecting key oil production drivers identified by ML and supported by process physics.

3. Methods

The methodological approach for this study followed a structured five-phase data science roadmap, as illustrated in Figure 1. This workflow encompasses the entire data lifecycle, from initial integrity checks to final model validation on unseen test data.

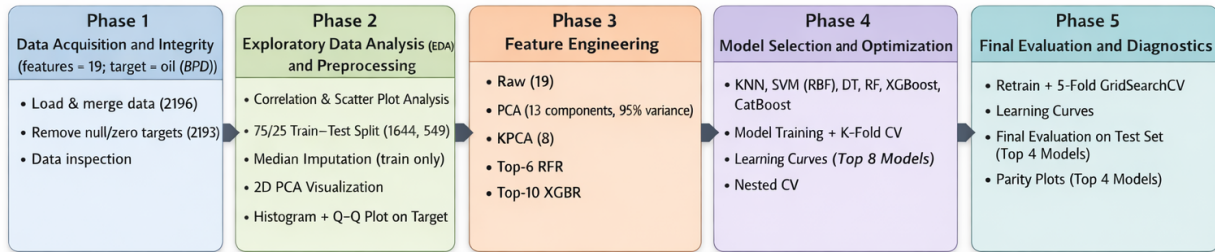


Figure 1: Methodological workflow: The 5-phase data Science roadmap ensures reproducible results and prevents overfitting when transforming wellhead data into production insights.

3.1. Phase 1 - Data Acquisition and Integrity

The study utilized a dataset of 2,196 observations from 22 Nigerian oil wells. The data was loaded using Pandas, treating commas as thousands separators to ensure numerical integrity, and entries with null or zero target values were removed. Zero production values correspond to non-producing or shut-in wells and are not representative of normal operating conditions. This resulted in 2,193 observations for modeling.

3.2. Phase 2 - Data Exploratory Analysis and Preprocessing

The data was split into 75% training (1,644 samples) and 25% testing (549 samples) sets, using a randomized split with a fixed seed to ensure reproducibility. To handle missing values, median imputation was applied to the training set and subsequently mapped to the test set to avoid data leakage. A correlation heatmap and scatter plots were used to identify linear and non-linear dependencies, while a 2D PCA plot was utilized for high-dimensional visualization to assess the structure and overlap of production levels in the feature space. To ensure that the variance-based PCA was not dominated by features with larger magnitudes, the features were standardized using a Z-score scaler (StandardScaler) to bring them onto a comparable scale. Finally, histograms and Q-Q plots were used to visualize the distribution of the oil (BPD) target.

Features	Minimum	Maximum	Mean	Median	Variance	Standard deviation
choke_64	12.0	137.00	74.35	70.0	1236.01	35.16
bsw_pct	0.0	97.33	42.18	47.0	968.36	31.12
fthp_psig	82.0	1665.70	316.22	259.3	37116.94	192.66
water_bpd	0.0	7268.00	1650.69	1301.6	2539896.43	1593.71
Oil (BPD)	30.8	10618.00	1859.35	1433.0	1867166.52	1366.44

Table 2: The train data summary statistics, showing the range, central tendency, and variability of key features and target.

Features	Minimum	Maximum	Mean	Median	Variance	Standard deviation
choke_64	10.0	137.0	73.54	70.0	1153.93	33.97
bsw_pct	0.0	95.0	40.20	46.0	934.09	30.56
fthp_psig	105.0	3255.0	329.09	265.7	51695.13	227.37
water_bpd	0.0	6500.0	1541.13	1202.2	2323848.72	1524.42
Oil (BPD)	9.0	10498.0	1992.42	1528.5	2263793.84	1504.59

Table 3: The test data summary statistics, showing the range, central tendency, and variability of key features and target.

3.3. Phase 3 - Feature Engineering

Five distinct feature sets, or "data kinds," were engineered to optimize the capture of complex wellhead physics. These included the Raw set of 19 initial features and a PCA set reduced to 13 components to explain 95.46% of the variance. Additionally, a nonlinear Kernel PCA (KPCA) set with 8 components was generated using an RBF kernel to capture high-dimensional patterns. Targeted feature selection was also performed using model-based importance rankings, creating a Top-6 RFR set from Random Forest and a Top-10 XGBR set from XGBoost Regressors.

As noted earlier, features used for PCA, KPCA, and the distance based nonlinear selected models (KNN and SVM-RBF) were standardized accordingly to ensure comparable feature scales. Conversely, tree-based and boosting models (Decision Tree, Random Forest, XGBoost, and CatBoost) were trained on unscaled data, as these algorithms are scale-invariant and do not require normalization for their splitting logic.

3.4. Phase 4 - Model Selection and Optimization

Six non-linear algorithms were evaluated across all five feature sets. These included KNN, SVM (RBF), Decision Trees, Random Forest, XGBoost, and CatBoost. The process utilized 10-fold Cross-Validation with shuffling enabled. This baseline ranking shortlisted the top eight model-data configurations, concentrating subsequent computational effort on the most promising architectures for the Nigerian wellhead data. For these finalists, learning curves were utilized to diagnose the bias-variance tradeoff. These diagnostics showed that the selected models were capturing generalizable relationships rather than memorizing noise.

3.5. Phase 5 - Final Evaluations and Diagnostics

This phase focused on hyperparameter optimization via Nested Cross-Validation and final model validation on the held-out test set. The eight shortlisted model-data configurations, comprising specific pairings of CatBoost, XGBoost, and Random Forest with engineered feature sets, were subjected to a rigorous 5x5 Nested Cross-Validation framework. This procedure utilized an inner

5-fold loop for exhaustive Grid Search hyperparameter tuning and an outer 5-fold loop to provide an unbiased estimate of generalization performance.

To maximize data utility, optimized models were retrained on the complete 75% training block (1,644 samples). The final predictive robustness was then validated against the held-out 25% test set (549 samples). Model performance was quantified using R^2 to assess explained variance, RMSE and MAE to measure absolute deviation in BPD units and Mean Absolute Percentage Deviation (MAPD) to evaluate relative error percentage. Additionally, the Standard Deviation of Residuals (STBD) was utilized to assess the consistency of the error distribution across the prediction range. Finally, parity and residual plots were generated for the top four candidates as the primary diagnostic tool to visually confirm forecast consistency across the entire production regime.

4. Results and Discussion

4.1. Exploratory Insights and Data Structure

The randomized 75/25 split ensured statistical consistency across subsets. As shown in Tables 2 and 3, key features like choke_64 and bsw_pct (Ref) exhibit nearly identical means and variance. This alignment confirms the training set is a reliable proxy for the underlying wellhead physics. As a result, identifying patterns in this consistent training block ensures exploratory insights represent the entire population rather than a biased split.

I. Correlation Heatmap and PCA Visualization of Production Categories

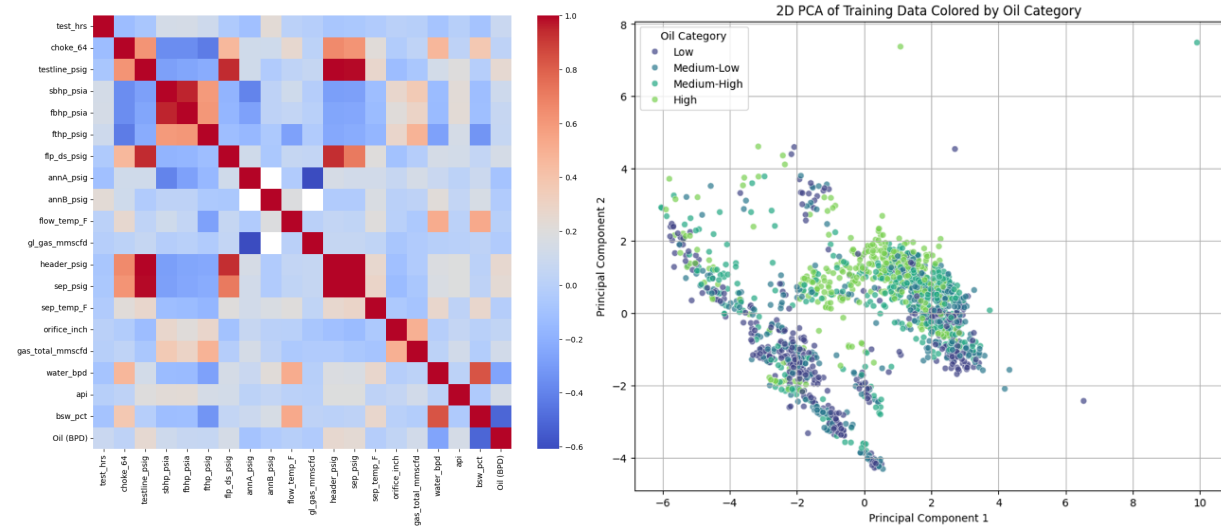


Figure 2: *Pearson Correlation Matrix (left) and 2D PCA projection (right) diagnosing the complex non-linear structure of the wellhead data*

The correlation matrix identifies bsw_pct (0.51) and water_bpd (0.26) as the strongest linear indicators, while primary physical drivers such as choke_64 and fthp_psig (Ref) exhibit near-zero linear correlation with oil production. This absence of linearity is mathematically validated by the 2D PCA projection, which shows a gradual overlap between production categories rather than sharp, linear separability. Together, these diagnostics made it evident that Nigerian oil production regimes exist on non-linear manifolds, showing that linear architectures would fail to resolve the system's physics and justifying the transition to advanced ensemble learning.

II. PCA Explained Variance and Component Selection

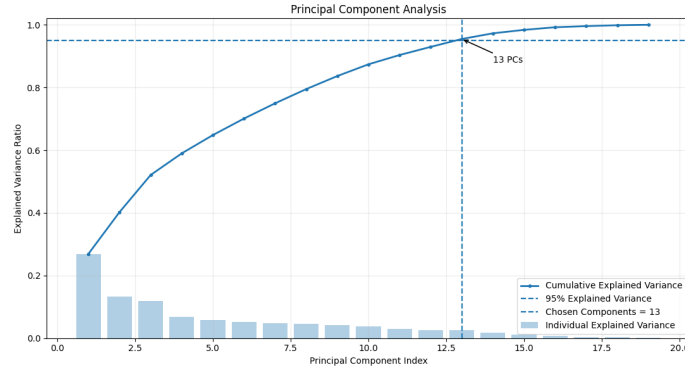


Figure 3: Cumulative explained variance ratio across principal components identifying the system's intrinsic dimensionality.

This reveals that 13 components capture 95.46% of the total variance, showing that the 19-dimensional wellhead space may contain significant linear redundancy. This mathematical compression reduces dimensionality by removing low-variance components, which may correspond to noise. Consequently, this 13-component subspace may improve computational efficiency for subsequent ensemble learning while preserving the dominant structure in the data.

III. Dominant Features Influencing Production

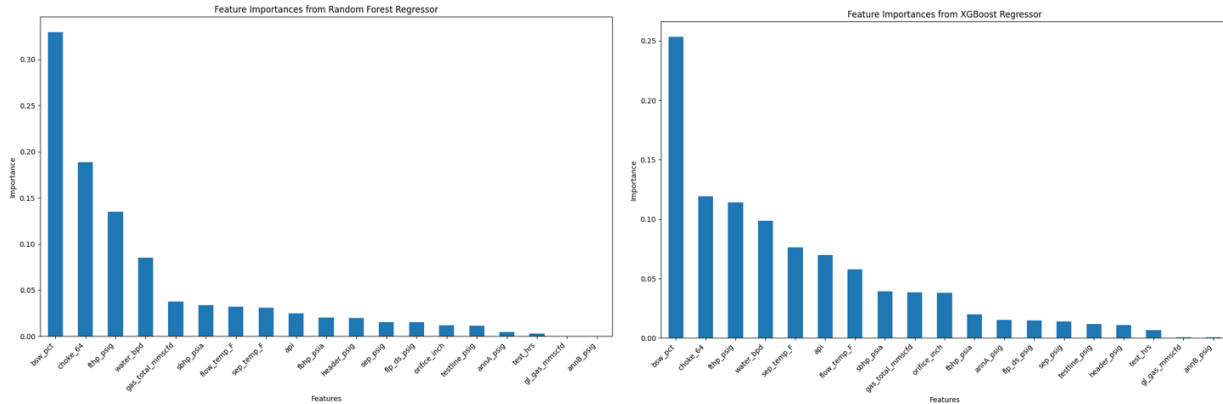


Figure 4: Global feature importance rankings from RFR (left) and XGBR (right) validating the physical drivers of oil production.

Both architectures exhibit strong algorithmic consensus, identifying `bsw_pct`, `choke_64`, `fthp_psig`, and `water_bpd` as the four primary predictors of production. This ranking is consistent with established literature and experimental observations [1][2]. It suggests that water cut, surface pressure, and choke size remain among the definitive constraints on multiphase production across diverse global oilfields. These rankings justified the creation of the Top-6 and Top-10 engineered feature sets used to train the final high-performance models.

4.2 Initial Model Screening and Shortlisting

To identify the most robust predictive frameworks, Phase 4 performed a comprehensive evaluation of 30 model-data combinations (6 algorithms paired with 5 engineered feature subsets). These baseline estimates served as a diagnostic filter, isolating algorithms with the mathematical depth required to capture the intricate multiphase flow dynamics characteristic of Nigerian oilfields.

	Data Kind	Model	R^2 mean	MAE mean	RMSE mean	MAPD mean	STBD mean
29	Top6RFR	CatBoost	0.901746	215.683047	419.878010	15.805135	419.610507
23	Top10XGR	CatBoost	0.899923	227.758442	422.875950	17.151119	421.757503
5	Raw	CatBoost	0.899129	231.697647	425.872148	17.523922	425.036386
28	Top6RFR	XGBoost	0.885942	235.553548	455.254197	16.639470	454.908307
21	Top10XGR	Random Forest	0.884018	231.656747	457.643516	15.954301	457.383564
27	Top6RFR	Random Forest	0.883626	224.466193	459.520196	14.780192	459.512179
22	Top10XGR	XGBoost	0.881694	245.253228	462.210331	18.375625	462.192312
4	Raw	XGBoost	0.876976	253.797570	470.982914	19.562802	470.964143

Table 4: Baseline ranking of the top eight model-data configurations based on 10-fold Cross-Validation performance metrics.

Initial screening reveals that CatBoost and XGBoost architectures dominate, with the Top6RFR_CatBoost configuration achieving a R^2 mean of 0.902 and a mean RMSE of 419.88 BPD. These top-tier performers demonstrated the mathematical depth required to approximate the high-dimensional flow regimes identified in the exploratory phase.

I. Diagnostic Curves of Non-Competitive Models

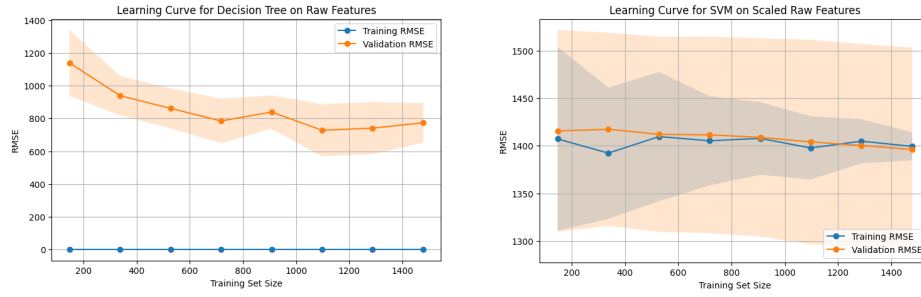


Figure 5: Diagnostic learning curves for Decision Tree (left) and SVM (right) illustrating the bracketing failure modes of the bias-variance trade-off.

The Decision Tree exhibits severe overfitting (memorizing noise) while the SVM suffers from underfitting (high bias), proving neither architecture can resolve the complex non-linear physics of the Nigerian wellhead data. These diagnostics mathematically justify excluding both models in favor of the balanced ensemble finalists presented in Figure 6.

II. Comparative Diagnostics of Shortlisted Ensembles

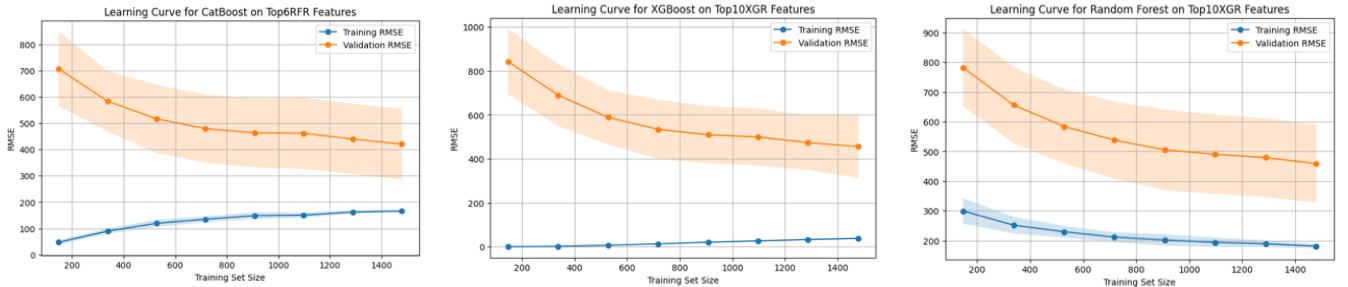


Figure 6: Comparative learning curves diagnosing the bias-variance signatures of the ensemble finalists.

CatBoost achieves peak predictive performance despite controlled overfitting, whereas XGBoost displays higher capacity and variance, reflected in a wider generalization gap. Random Forest exhibits the most stable balance between bias and variance. The downward trajectory of validation

error across all models confirms that the observed gaps would continue to close with additional training data. This diagnostic trend proves the architectures successfully learned the complex wellhead physics rather than memorizing training noise.

4.3 Final Model Validation

Phase 5 executed the final evaluation on the unseen test set (549 samples) after a rigorous hyperparameter optimization stage, benchmarking real-world forecasting reliability. This stage confirms whether the architectures, once locked into their optimal configurations, successfully generalized the complex physics identified during training.

I. Final Test Set Performance and Model Ranking

Tag	Data kind	Model	R^2	MAE	RMSE	MADP	STBD	Rank
t10xgr_cat	Top10XGR	CatBoost	0.903391	248.681017	467.231572	37.811085	371.912177	1
t6rfr_cat	Top6RFR	CatBoost	0.898858	245.942256	478.066137	34.619465	324.395151	2
t10xgr_xgb	Top10XGR	XGBoost	0.888373	260.169714	502.235307	45.262598	460.198758	3
t6rfr_xgb	Top6RFR	XGBoost	0.885561	255.326913	508.522818	41.395990	420.311829	4
raw_xgb	Raw	XGBoost	0.884052	249.659902	511.863732	27.029994	174.617224	5
raw_cat	Raw	CatBoost	0.883004	267.077131	514.170675	35.637599	302.439192	6
t10xgr_rf	Top10XGR	Random Forest	0.875294	263.526703	530.843747	33.220233	263.762880	7
t6rfr_rf	Top6RFR	Random Forest	0.874602	257.858642	532.314485	34.892909	314.919929	8

Table 5: Final test set performance and ranking of the top eight model-data configurations. Results demonstrate strong consistency with cross-validation rankings, confirming a good and stable model selection process.

CatBoost frameworks dominate the top tier, led by the t10xgr_cat configuration with peak generalization ($R^2 = 0.903$; RMSE = 467.23 BPD). Notably, XGBoost variants benefited significantly from the final hyperparameter tuning phase, effectively narrowing their generalization gaps to secure the #3 and #4 ranks. Conversely, Random Forest models show comparatively lower performance on the test set, which may be due to higher sensitivity to unseen data distributions.

II. Optimized Model Configurations

The table below summarizes the best hyperparameter settings obtained after model tuning for each shortlisted configuration.

Tag	Data Kind	Model	Best hyperparameters
raw_xgb	Raw	XGBoost	{'colsample_bytree': 1.0, 'learning_rate': 0.03, 'max_depth': 7, 'n_estimators': 5000, 'subsample': 0.8}
t6rfr_cat	Top6RFR	CatBoost	{'depth': 8, 'iterations': 5000, 'l2_leaf_reg': 5, 'learning_rate': 0.03}
raw_cat	Raw	CatBoost	{'depth': 8, 'iterations': 5000, 'l2_leaf_reg': 3, 'learning_rate': 0.05}
t10xgr_cat	Top10XGR	CatBoost	{'depth': 8, 'iterations': 5000, 'l2_leaf_reg': 5, 'learning_rate': 0.05}
t10xgr_xgb	Top10XGR	XGBoost	{'colsample_bytree': 0.8, 'learning_rate': 0.05, 'max_depth': 7, 'n_estimators': 5000, 'subsample': 0.8}
t6rfr_xgb	Top6RFR	XGBoost	{'colsample_bytree': 0.8, 'learning_rate': 0.03, 'max_depth': 7, 'n_estimators': 5000, 'subsample': 0.8}
t10xgr_rf	Top10XGR	Random Forest	{'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300}
t6rfr_rf	Top6RFR	Random Forest	{'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 800}

Table 6: Optimal hyperparameter configurations for the top eight model-data combinations.

The optimized hyperparameters show that boosting models rely on deeper trees and smaller learning rates, while Random Forest uses moderate depth and ensemble size to maintain stability.

III. Visual Diagnostics of Model Performance

Parity and residual plots are used to evaluate prediction accuracy and assess error behavior across the production range.

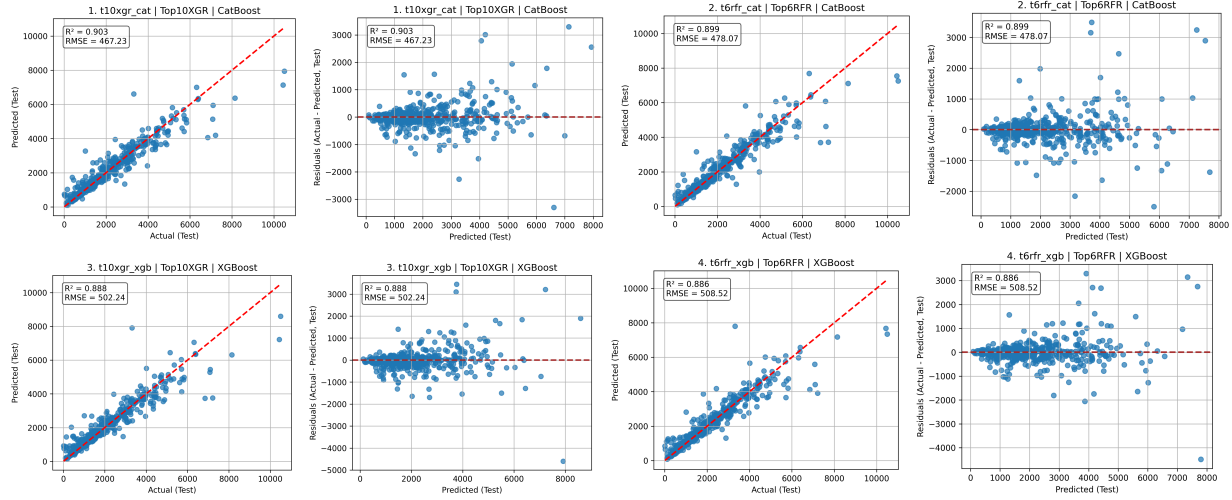


Figure 7: Parity plots for the top four models show strong agreement between predicted and actual values, with points clustering near the unit-slope line across the production range. Residual plots are centered around zero with no clear pattern, indicating stable errors and minimal systematic bias.

The consistent model performance across both high and low production ranges validates that the engineered feature subsets successfully captured the underlying physical signal of the Nigerian wellhead data.

5. Conclusion and Prospects

This study developed a robust predictive framework for oil production using non-linear ensemble models. CatBoost achieved the best performance ($R^2 = 0.903$) with the lowest RMSE, effectively modeling relationships where conventional correlations fail. Feature analysis identified `bsw_pct`, `choke_64`, `fthp_psig`, and `water_bpd` as the dominant drivers, aligning with the process physics. Targeted feature selection outperformed PCA, KPCA, and raw datasets in generalization. Boosting models delivered the highest accuracy, while Random Forest balanced stability with performance, as observed in its learning curve.

Overall, the framework provides a high-accuracy tool for production prediction, with potential applications including predicting performance for wells with no production history. Future work will focus on expanding the dataset to leverage the non-plateaued learning curves and improve model precision. It will also include well-to-well analysis across broader Nigerian oil fields to enhance generalization and understanding of production dynamics.

6. References

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